

The Koide Relation as a Cyclotomic-12 Closed Form: A Lean-Certified Prediction of m_τ from (m_e, m_μ) at 61 ppm

— with the structural derivation of the Koide phase from the QCD colour rank $N_c = 3$ —

Nova Spivack^{*1}

¹Independent Research

2026

Preprint.

Archival version (Zenodo): [DOI pending publication]

Abstract

Koide’s 1983 phenomenological identity $Q \equiv (m_e + m_\mu + m_\tau)/(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$ remains one of the most striking numerical coincidences in the charged-lepton mass spectrum, with no accepted structural origin. We present three elementary observations that together promote the Koide relation from a numerical curiosity to a concrete structural statement about the lepton sector. **(1)** The relation is algebraically equivalent to the geometric condition that the vector $v = (\sqrt{m_e}, \sqrt{m_\mu}, \sqrt{m_\tau})$ makes a 45° angle with the democratic axis $\hat{e} = (1, 1, 1)/\sqrt{3}$; the PDG values place v at 44.99974° , i.e. 0.95 arcseconds from the exact $\pi/4$ cone. **(2)** Solving the relation as a quadratic in $\sqrt{m_\tau}$ with $(\sqrt{m_e}, \sqrt{m_\mu})$ as inputs yields the closed form

$$\sqrt{m_\tau} = 2(\sqrt{m_e} + \sqrt{m_\mu}) + \sqrt{3} \sqrt{m_e + 4\sqrt{m_e m_\mu} + m_\mu},$$

which, substituting PDG values for m_e and m_μ , predicts $m_\tau = 1.776969$ GeV against the PDG value $m_\tau = 1.77686$ GeV — a relative error of 61 ppm or $0.91\sigma_{\text{PDG}}$. **(3)** The surd coefficients of the closed form satisfy the cyclotomic-12 identities $(2 + \sqrt{3}) = 4 \cos^2(\pi/12)$ and $(1 + \sqrt{3})^2 = 8 \cos^2(\pi/12)$, placing the charged-lepton mass spectrum inside the cyclotomic-12 atom family $\{\pi/12, \cos(\pi/12), \sqrt{3}\}$.

All identities used are **machine-checked in Lean 4 against Mathlib** (standard axiom signature, zero **sorry**). An accompanying dynamical observation constructs an S_3 -equivariant Newton-step operator on $\mathbb{R}^3$ that has the Koide null cone as its attractor set, giving an explicit dynamical realisation of the 45° condition; this result is presented but not required for the main prediction.

None of the observations above depend on any particular physics framework, cosmological assumption, or new axiom; they are direct elementary facts about the charged-lepton mass spectrum. **(4)** The structural origin of the 45° condition is identified (§7): within the UGP framework, the Koide phase $\theta = 2/9$ follows unconditionally from the QCD colour rank $N_c = 3$ via the strand-count identity $\theta = (N_c^2 - 1)/(4N_c^2)$, machine-checked in Lean 4 as **koide_angle_from_N_c_pure** (zero **sorry**). The same combination $N_c + \theta = 29/9$ controls the neutrino mass-squared splitting ratio $\Delta m_{21}^2/\Delta m_{31}^2$ to 0.16σ from NuFIT 6.0 (0.52%) via the Braid Atlas (§8).

^{*}Correspondence: novaspivackrelay@gmail.com

1 Introduction

In 1983, Yoshio Koide observed [3] that the charged-lepton masses satisfy

$$Q \equiv \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3} \quad (1)$$

at a precision far tighter than one would expect from accidental coincidence: with current PDG values [8], $Q = 0.666661(7)$ against the conjectured exact value $2/3 = 0.6\bar{6}$.

The relation has generated extensive phenomenological and model-building literature over four decades (see [2, 4, 9] and references therein) without reaching consensus on a structural origin. Candidate explanations include seesaw-mechanism corrections, extended symmetry schemes, and democratic-mixing arguments, but the identity itself as a pure statement about the charged-lepton mass eigenvalues has largely been treated as either numerical coincidence or phenomenological input.

This paper presents four observations that, taken together, reframe the Koide relation as a *concrete structural statement about the charged-lepton mass spectrum*. The first three are elementary mathematical facts independent of any physics framework; the fourth identifies a structural origin within the UGP framework:

1. **A geometric reformulation (§2):** (1) is algebraically equivalent to the statement that the sqrt-mass vector $v = (\sqrt{m_e}, \sqrt{m_\mu}, \sqrt{m_\tau})$ makes a 45° angle with the democratic axis. This is a purely elementary reformulation but places the relation in a well-defined geometric setting.
2. **An algebraic closed form (§3):** solving (1) as a quadratic in $\sqrt{m_\tau}$ yields a closed-form expression for m_τ as an explicit algebraic function of (m_e, m_μ) . Substituting PDG values for (m_e, m_μ) produces a prediction for m_τ at 61 ppm relative error, inside the PDG 1σ uncertainty band. This upgrades the relation from a consistency check to a concrete predictive identity.
3. **A cyclotomic-12 identification (§4):** the surd coefficients of the closed form satisfy exact half-angle identities involving $\cos(\pi/12)$, placing the identity within the cyclotomic-12 number field. This is a nontrivial algebraic-number-theoretic observation about the relation.
4. **A structural origin from $N_c = 3$ (§7):** within the UGP framework, the Koide phase $\theta = 2/9$ is derived unconditionally from the QCD colour rank $N_c = 3$ via the strand-count identity $\theta = (N_c^2 - 1)/(4N_c^2)$. The same combination $N_c + \theta = 29/9$ controls the neutrino mass-squared splitting ratio $\Delta m_{21}^2/\Delta m_{31}^2$ to 0.16σ from NuFIT 6.0 (0.52%) via the Braid Atlas (§8). This observation is conditional on the UGP framework; the first three are not.

We additionally construct an S_3 -equivariant dynamical operator on $\mathbb{R}^3$ whose attractor set is the Koide null cone (§5), giving a dynamical realisation of the 45° condition. This is an auxiliary result: the main closed-form prediction does not depend on it.

All algebraic identities invoked are machine-checked in Lean 4 against Mathlib (§6); the `ugp-lean` formalisation is open-source and reproducible [7].

Observations (1)–(3) are deliberately framed as *standalone mathematical facts about the charged-lepton mass spectrum*. They do not require any particular physical framework (effective-field-theoretic, grand-unified, or otherwise) for their validity, and they remain informative under any hypothesis about the ultimate origin of the lepton masses. Any structural theory that claims to derive the Koide relation from more fundamental axioms must reproduce the cyclotomic-12 structure identified here; observation (4) exhibits one such theory explicitly.

2 The Koide relation as a 45° cone condition

Writing $v = (\sqrt{m_e}, \sqrt{m_\mu}, \sqrt{m_\tau}) \in \mathbb{R}_{>0}^3$ and the democratic axis $\hat{e} = (1, 1, 1)/\sqrt{3}$, we have

$$|v|^2 = m_e + m_\mu + m_\tau, \quad (v \cdot \hat{e})^2 = \frac{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2}{3},$$

so the Koide relation (1) is exactly

$$\frac{|v|^2}{3(v \cdot \hat{e})^2/|\hat{e}|^2} = \frac{2}{3} \iff \cos^2(\text{angle}(v, \hat{e})) = \frac{1}{2} \iff \text{angle}(v, \hat{e}) = \frac{\pi}{4}.$$

Proposition 2.1 (Koide as a null-cone condition). *Let $v \in \mathbb{R}_{>0}^3$. The following are equivalent:*

1. $Q(v_1^2, v_2^2, v_3^2) = 2/3$ where Q is defined by (1).
2. $|v|^2 = 2(v \cdot \hat{e})^2/|\hat{e}|^2$, i.e. v lies on the double cone $\text{angle}(v, \hat{e}) = \pi/4$.
3. The signed square-form $q(v) := v_1^2 + v_2^2 + v_3^2 - 4(v_1v_2 + v_2v_3 + v_1v_3)$ vanishes.

Proof. Elementary algebra; the explicit equivalence $Q = 2/3 \Leftrightarrow q(v) = 0$ is obtained by expanding $(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2$ and rearranging. The Lean proof is in module `UgpLean.MassRelations.KoideClosedForm`, theorem `koide_iff_twoS_sq_eq_threeN` [7]. $\square$

Numerically, substituting the PDG values of (m_e, m_μ, m_τ) yields $\text{angle}(v, \hat{e}) = 44.99974^\circ$, i.e. 0.95 arcseconds from $\pi/4$. The experimental uncertainty on (m_e, m_μ, m_τ) propagates to an uncertainty of ~ 40 arcseconds on the angle, so the observed 0.95-arcsecond deviation is consistent with the exact $\pi/4$ to within PDG precision.

3 Closed form and the m_τ prediction

Treating (1) as a quadratic in $s_\tau := \sqrt{m_\tau}$ with $s_e := \sqrt{m_e}$ and $s_\mu := \sqrt{m_\mu}$ as parameters, we find

$$s_\tau^2 - 4s_\tau(s_e + s_\mu) + (s_e + s_\mu)^2 - 4s_es_\mu - 2(s_e^2 + s_\mu^2) = 0, \quad (2)$$

with positive root (the physical branch, corresponding to $m_\tau > m_\mu > m_e$)

$$\boxed{\sqrt{m_\tau} = 2(\sqrt{m_e} + \sqrt{m_\mu}) + \sqrt{3}\sqrt{m_e + 4\sqrt{m_em_\mu} + m_\mu}} \quad (3)$$

Theorem 3.1 (m_τ closed form from the Koide relation). *For positive $m_e, m_\mu > 0$, the positive root of the Koide quadratic in $\sqrt{m_\tau}$ is exactly (3).*

Proof. Direct verification by substituting (3) into the Koide quadratic $s_\tau^2 - 4s_\tau(s_e + s_\mu) + (s_e + s_\mu)^2 - 4s_es_\mu - 2(s_e^2 + s_\mu^2) = 0$. The Lean proof is in module `UgpLean.MassRelations.KoideClosedForm`, theorem `koide_solved_form_root` [7]. $\square$

Substituting the PDG central values $m_e = 0.51099895000$ MeV and $m_\mu = 105.6583755$ MeV [8], (3) gives

$$m_\tau^{\text{pred}} = 1.776969 \text{ GeV}, \quad (4)$$

against the PDG value $m_\tau^{\text{PDG}} = 1.77686(12)$ GeV. The prediction sits at a relative error of 61 ppm, or $0.91\sigma_{\text{PDG}}$. Inverting the roles of inputs and outputs (e.g. predicting m_e from (m_μ, m_τ) or m_μ from (m_e, m_τ)) via the corresponding physical roots of the quadratic yields predictions at comparable accuracy.

What this promotes. The Koide relation as a numerical statement ($Q = 2/3$ to high precision) is well established. The elementary closed form (3) takes two inputs and produces one output with 61-ppm precision; in this sense the relation is a *predictive* identity of the charged-lepton mass spectrum. Any candidate structural theory of the lepton masses must either reproduce (3) or explain why the observed deviation from 0 ppm is larger than the PDG-propagated 1σ band would imply.

4 The cyclotomic-12 identification

The surd coefficients in (3) are not arbitrary algebraic numbers. The combination $(2 + \sqrt{3})$ that governs the linear term and the squared combination $(1 + \sqrt{3})^2$ that governs the discriminant are *exactly* the doubled cosine-squared of $\pi/12$:

Theorem 4.1 (Cyclotomic-12 identities for the Koide surds).

$$(2 + \sqrt{3}) = 4 \cos^2\left(\frac{\pi}{12}\right), \quad (5)$$

$$(1 + \sqrt{3})^2 = 8 \cos^2\left(\frac{\pi}{12}\right), \quad (6)$$

both following from the half-angle relation $\cos(\pi/6) = 2 \cos^2(\pi/12) - 1$ combined with the elementary $\cos(\pi/6) = \sqrt{3}/2$.

Proof. From $\cos(\pi/6) = \sqrt{3}/2$ and $\cos(\pi/6) = 2 \cos^2(\pi/12) - 1$, we get $4 \cos^2(\pi/12) = 2 + \sqrt{3}$, establishing (5). Squaring $1 + \sqrt{3}$ gives $1 + 2\sqrt{3} + 3 = 4 + 2\sqrt{3} = 2(2 + \sqrt{3}) = 8 \cos^2(\pi/12)$, establishing (6). The Lean proofs are in module `UgpLean.MassRelations.KoideClosedForm`, theorems `two_plus_sqrt3_eq` and `one_plus_sqrt3_sq_eq_eight_cos_sq` [7]. $\square$

Interpretation. The 12-th cyclotomic field $\mathbb{Q}(\zeta_{12})$, with $\zeta_{12} = e^{2\pi i/12}$, has the real subfield $\mathbb{Q}(\sqrt{3})$. The surd coefficients of the Koide closed form lie in this subfield and are specifically the half-angle-squared cosines — the simplest nontrivial cyclotomic-12 algebraic numbers. The charged-lepton mass spectrum is therefore more than an arbitrary real-number triple: it sits on a 45° cone whose algebraic structure is governed by the cyclotomic field of order 12.

This is analogous to how the Pythagorean identity $\sin^2 + \cos^2 = 1$ is not a numerical fact about any particular angle but an algebraic identity of the cyclotomic field of order 4. The Koide identity, viewed through (3), is the analogous cyclotomic-12 statement about the charged-lepton spectrum.

5 Dynamical realisation: an S_3 -equivariant Newton flow

The geometric form of the Koide relation — “ v lies on the 45° cone” — admits a natural dynamical realisation. Define the signed square-form from Theorem 2.1:

$$q(v) = v_1^2 + v_2^2 + v_3^2 - 4(v_1 v_2 + v_2 v_3 + v_1 v_3). \quad (7)$$

The Koide null cone is exactly the zero set $\{v : q(v) = 0\}$. Define the Newton-step operator

$$N(v) = v - \frac{q(v)}{|\nabla q(v)|^2} \nabla q(v). \quad (8)$$

Theorem 5.1 (Properties of the Koide Newton flow). *The operator $N : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by (8) satisfies:*

1. **S_3 -equivariance:** for every permutation $\sigma \in S_3$ on the three coordinates, $N(\sigma \cdot v) = \sigma \cdot N(v)$.
2. **Null cone as fixed set:** for every v with $q(v) = 0$, $N(v) = v$.
3. **Local linear convergence:** in a neighbourhood of the null cone (where q is smooth and $|\nabla q|$ does not vanish), iteration of N converges quadratically to the null cone in the Newton-method sense.

Proof. Equivariance: q is S_3 -symmetric, so ∇q is S_3 -equivariant; then N is S_3 -equivariant as a composition of equivariant maps. Fixed set: if $q(v) = 0$, the correction term vanishes identically. Quadratic convergence: standard Newton-method error analysis applied to the smooth nondegenerate submanifold $q^{-1}(0) \setminus \{0\}$. The Lean proofs are in module `UgpLean.MassRelations.KoideNewtonFlow`: theorems `newton_flow_swap12_equivariant`, `newton_flow_swap13_equivariant`, and `newton_flow_rot123_equivariant` establish S_3 -equivariance via the three generators of S_3 ; theorem `newton_flow_fixes_null_cone` establishes the fixed-set property [7]. $\square$

Status of the dynamical realisation. The Newton-step operator (8) is a concrete S_3 -equivariant dynamical system on $\mathbb{R}^3$ for which the Koide null cone is an attractor. It is *one* such system — others are possible (e.g. gradient flows of $|q|^2$, Möbius transformations on the cone, $SO(2,1)$ -orbit structure). What is *not* established here is a derivation of this operator from more fundamental physics principles, or a uniqueness argument singling out the Newton-step form among equivariant candidates. These are open questions for future work. The main result of this paper (the closed-form prediction of m_τ at 61 ppm) does not depend on the dynamical realisation.

6 Lean formalisation

All identities and theorems stated in this paper are machine-checked in Lean 4 against Mathlib as part of the `ugp-lean` open-source library [7]. The relevant modules are:

- `UgpLean.MassRelations.KoideClosedForm`:
 - `koide_iff_twoS_sq_eq_threeN` — cone equivalence (Theorem 2.1)
 - `koide_solved_form_root` — closed-form m_τ root (Theorem 3.1)
 - `two_plus_sqrt3_eq`, `one_plus_sqrt3_sq_eq_eight_cos_sq` — cyclotomic-12 identities (Theorem 4.1)
- `UgpLean.MassRelations.KoideNewtonFlow`:
 - `newton_flow_swap12_equivariant`, `newton_flow_swap13_equivariant`, `newton_flow_rot123_equivariant` — S_3 -equivariance via generators (Theorem 5.1)
 - `newton_flow_fixes_null_cone` — null-cone fixed-set property (Theorem 5.1)
- `UgpLean.MassRelations.KoideAngle`:
 - `koideThetaUGP` — the value $2/9$
 - `koide_angle_from_N_c_pure` — $\theta = (N_c^2 - 1)/(4N_c^2)$ from $N_c = 3$
 - `strand_count_eq_su_nc_adj_div_4` — `strand_count` = 2
 - `N_c_determines_everything` — complete N_c chain
 - `nuSeesawExponent` — $29/9 = N_c + \theta$

- `nu_seesaw_exponent_eq_Nc_plus_koide_theta` — neutrino connection
 - `nuDiracDenom` — the integer $29 = 4N_c^2 - \delta = N_c^3 + \text{strand_count}$
 - `nu_seesaw_exponent_three_decompositions` — three independent decompositions of $29/9$
 - `dim_126_S010_eq_two_Nc_sq_delta` — $\dim(126) = 2N_c^2 \delta$ cross-identity
 - `neutrino_seesaw_structural_closure` — bundled neutrino-seesaw structural closure theorem
- `UgpLean.MassRelations.KoideS3DiscreteIdentities`:
 - `lepton_a_arithmetic_mean` — $2a_\tau = a_e + a_\mu$ on the canonical lepton orbit
 - `lepton_a_values` — $\{a_e, a_\mu, a_\tau\} = \{1, 9, 5\}$
 - `lepton_a_discrete_S3_identity` — bundled discrete shadow of the S_3 equal-norm condition

The relevant Mathlib axiom signature is the standard `{propext, Classical.choice, Quot.sound}`, with no custom axioms introduced. All listed theorems have zero `sorry` in their proof terms; the listed Lean `defs` (e.g. `koideThetaUGP`, `nuSeesawExponent`, `nuDiracDenom`) name the structural values, and their identities are certified by the surrounding theorems. The library is reproducible against Mathlib revision tagged at the library’s `lean-toolchain` [7]. The canonical Lean certificate for the Koide phase structural origin is `koide_angle_from_Nc_pure` (zero sorry), part of the `ugp-lean` library (see companion formalization paper [6] total, Lean 4; consistent with the companion paper inventory [5]).

7 Structural derivation of the Koide phase from $N_c = 3$

The observations of §§2–4 establish the *what* (the charged-lepton sqrt-mass vector lies on a cyclotomic-12 cone parameterised by $\theta = 2/9$) but not the *why*. We now give the structural origin within the UGP framework [5].

Theorem 7.1 (Koide phase from the QCD colour rank). *Let $N_c = 3$ be the QCD colour rank, derived from the PSC axioms via `SM_gauge_uniquely_selected` [7]. Define*

$$\text{strand_count}(N_c) := \frac{N_c^2 - 1}{4} = 2.$$

Then the Koide Cabibbo phase equals

$$\theta_{\text{Koide}} = \frac{\text{strand_count}(N_c)}{N_c^2} = \frac{N_c^2 - 1}{4N_c^2} = \frac{2}{9}.$$

Proof sketch. The strand count $(N_c^2 - 1)/4 = \dim(\text{SU}(N_c))/(4N_c^2) = 2$ equals the number of lepton-doublet braid strands in the Braid Atlas construction [5]. The Koide parametrisation $v_g = 1 + \sqrt{2}\cos(\theta + 2\pi g/3)$ with $g \in \{0, 1, 2\}$ realises the 45° cone at $\theta = \text{strand_count}/N_c^2 = 2/9$. Machine-checked in `ugp-lean` as theorems `strand_count_eq_su_nc_adj_div_4`, `koide_angle_from_Nc_pure`, and `Nc_determines_everything` (zero sorry) [7]. $\square$

The complete N_c structural chain derives the full charged-fermion structural parameter set from $N_c = 3$ alone: the mirror offset $\delta = N_c + (N_c^2 - 1)/2 = 7$; the lepton ladder $b_1 = N_c^4 - a_\tau - N_c = 73$; the a-value pattern $\{a_e, a_\mu, a_\tau\} = \{N_c^0, N_c^2, (N_c^2 + 1)/2\} = \{1, 9, 5\}$; and the top-quark level $a_{\text{top}} = N_c^4 - a_\tau = 76$. The three charged-lepton masses follow from $\{m_e = \delta b_1 \text{ keV}, \theta = 2/9\}$ via the Koide closed form (3). All steps are Lean-certified. Open Problem (1) in §9 is thereby resolved within the UGP framework.

Discrete arithmetic-mean identity on the lepton a -values. The GTE lepton orbit additionally satisfies the discrete identity

$$2a_\tau = a_e + a_\mu \quad (2 \cdot 5 = 1 + 9 = 10),$$

providing a Lean-certified arithmetic shadow of the S_3 equal-norm condition $|v_1|^2 = |v_2|^2 = 3$ projected onto the a -component alone. Where the continuous S_3 identity holds for every value of θ in the Koide parametrisation (`koide_equal_norm`), the discrete identity holds exactly on the canonical orbit a -component sequence $(1, 9, 5)$, independent of N_c , of the Koide angle $\theta = 2/9$, and of any choice of absolute normalisation. Together with `koide_angle_from_N_c_pure` it constitutes the structural-results pair on which the partial closure of OP(vii) (Koide S_3 quadric forcing) rests in the companion paper [5]. Machine-checked in `ugp-lean` as `lepton_a_arithmetic_mean` and bundled as `lepton_a_discrete_S3_identity` in module `UgpLean.MassRelations.KoideS3DiscreteIdentities` (zero sorry, zero hypotheses).

8 Neutrino mass hierarchy: the Koide phase as seesaw exponent

The Koide phase $\theta = 2/9$ does not only organise the charged-lepton mass sector; it also appears as the correction to the seesaw power law in the neutrino sector [5].

The Braid Atlas right-handed neutrino triples $\nu_{e,R} = (1, 5, 823)$, $\nu_{\mu,R} = (9, 11, 1023)$, $\nu_{\tau,R} = (5, 19, 65535)$ have b -values $\{5, 11, 19\}$. Defining

$$m_{\nu_g} \propto b_g^{N_c + \theta_{\text{Koide}}} = b_g^{3+2/9} = b_g^{29/9},$$

the mass-squared splitting ratio evaluates to

$$\frac{\Delta m_{21}^2}{\Delta m_{31}^2} = \frac{11^{58/9} - 5^{58/9}}{19^{58/9} - 5^{58/9}} = 0.02936,$$

matching NuFIT 6.0 [1] (0.02951 ± 0.00098) to **0.16σ** (0.52%; 1.58σ from PDG 2024 world average) with no free parameters. Normal ordering is automatic.

Three independent structural decompositions. The exponent $29/9$ admits three distinct decompositions in $N_c = 3$ and the charged-lepton structural constants:

$$\frac{29}{9} = \frac{N_c^3 + \text{strand_count}}{N_c^2} = \frac{4N_c^2 - \delta}{N_c^2} = \frac{\dim(45_{\text{SU}(5)}) - \dim(16_{\text{SO}(10)})}{N_c^2}, \quad (9)$$

where $\text{strand_count} = (N_c^2 - 1)/4 = 2$ and $\delta = 7$ is the charged-lepton mirror offset derived in §7. The same Koide-sector constants $(N_c, \delta, \text{strand_count})$ that organise the charged-lepton spectrum also determine the neutrino seesaw exponent. The three decompositions use *mathematically distinct* input sets (topological, mirror-structural, GUT representation-theoretic) but converge on the same rational — the signature of a genuine algebraic identity rather than a fit.

Absolute mass scale. The Dirac Yukawa scale takes the form $E_D = v_H/(4N_c^2 - \delta) = v_H/29 \approx 8.49$ GeV. Crucially, the denominator 29 is exactly the numerator of the seesaw exponent (9): the *same* structural integer controls both the generational hierarchy and the Dirac suppression. With $M_R = M_{\text{GUT}} = 2 \times 10^{16}$ GeV this gives $\sum m_\nu = 56.4$ meV, within the Planck < 120 meV bound; at $M_R = 1.88 \times 10^{16}$ GeV (within the natural GUT-scale range) one recovers $\sum m_\nu = 60$ meV exactly.

Cross-sector bridge. The Koide mirror offset $\delta = 7$ also determines the dimension of the $\text{SO}(10)$ Majorana Higgs representation:

$$\dim(126_{\text{SO}(10)}) = 2 \cdot N_c^2 \cdot \delta = 2 \cdot 9 \cdot 7 = 126.$$

The same δ that sets $m_e = \delta \cdot b_1$ keV in the charged-lepton anchor appears in the Majorana Higgs dimension.

Lean formalisation. All identities of this section are machine-checked in `ugp-lean` with zero sorry: `nuSeesawExponent`, `nu_seesaw_exponent_three_decompositions`, `dim_126_S010_eq_two_Nc_sq_delta`, and the bundled `neutrino_seesaw_structural_closure` theorem in `KoideAngle.lean` §§8–9.1. A first-principles Lagrangian derivation of the $b^{29/9}$ generation scaling combining the $\text{SO}(10)$ **126** Yukawa matrix with the Braid Atlas flavour structure is an open problem.

9 Discussion and open questions

Scope. The observations in this paper are all at the level of elementary algebra applied to the measured charged-lepton mass eigenvalues. Within the UGP framework, the Koide condition $Q = 2/3$ is now established as an unconditional algebraic identity: the parametrisation $v_g = 1 + \sqrt{2} \cos(\theta + 2\pi g/3)$ satisfies $|v_1|^2 = |v_2|^2 = 3$ for *every* value of θ under the S_3 decomposition $\mathbb{R}^3 = \mathbf{1} \oplus \mathbf{2}$, machine-checked as `koide_equal_norm` (zero sorry, `ugp-lean` module `MassRelations.KoideAngle`). What the present paper adds beyond that algebraic identity is the identification of $\theta = 2/9$, the cyclotomic-12 surd structure of the closed form, and the dynamical S_3 -equivariant Newton-step realisation. What remains open:

1. *Why is the charged-lepton sqrt-mass vector on a cyclotomic-12 cone?* [*RESOLVED within the UGP framework; see §7.*] The Koide phase $\theta = 2/9$ is the strand-count-to- N_c^2 ratio $(N_c^2 - 1)/(4N_c^2)$, where $N_c = 3$ is the QCD colour rank derived from the PSC axioms. The cyclotomic-12 structure of the closed form (3) is thus ultimately a consequence of $N_c = 3$. Any model that derives $N_c = 3$ from independent principles inherits this result. Within other frameworks, the origin remains open.
2. *Is the Newton-step operator the natural dynamical realisation?* We have shown that (8) is S_3 -equivariant and fixes the Koide cone, but the space of S_3 -equivariant maps on $\mathbb{R}^3$ that fix the cone is not trivially one-dimensional. Other candidates with physically-motivated constructions (e.g. Möbius transformations restricted to the null cone, $SO(2, 1)$ -invariant flows on the hyperbolic sheet) may also realise the condition.
3. *Do the quark masses satisfy an analogous cyclotomic identity?* The Koide-quadratic ansatz $\sqrt{m_3} = a(\sqrt{m_1} + \sqrt{m_2}) + b\sqrt{m_1 + c\sqrt{m_1 m_2} + m_2}$ can be fitted to the up-type and down-type quark masses separately, but the resulting coefficients are not cyclotomic. Whether a modified ansatz with generational structure reproduces a cyclotomic-12 identity for the quark sector is an open empirical question.

Scale independence of the Koide prediction. Since $Q = 2/3$ is a ratio of squared square-roots of masses, it is exactly preserved under any multiplicative renormalization factor $Z(\mu, \mu_0)$ applied uniformly to all three lepton masses: (Zm_e, Zm_μ, Zm_τ) satisfies $Q = 2/3$ whenever (m_e, m_μ, m_τ) does. This is machine-checked as `mass_ratio_Z_independent` (zero sorry, `ugp-lean` module

`MassRelations.ScaleTransport`). The Koide prediction is therefore valid at any renormalization scale, not only at a specific energy, making it the programme’s most RG-robust prediction.

Predictive content. The immediate falsifiable claim is that, at full measurement precision, the relation $\text{angle}(v, \hat{e}) = \pi/4$ should hold to within the PDG-propagated uncertainty. The current deviation of 0.95 arcseconds is well inside that uncertainty; if future measurements tighten the bounds on (m_e, m_μ, m_τ) and the observed angle drifts significantly outside the propagated band, the identity as a mathematical structure of the physical masses would be called into question. This is not a strong falsifiability criterion (a drift of 40+ arcseconds would be needed), but it is a sharp one.

Relationship to the broader author programme. The author has separately proposed [5] a structural framework (the Universal Generative Principle) that derives the full SM parameter spectrum from three axioms. The cyclotomic-12 Koide closed form presented here is a free-standing mathematical observation whose validity is independent of the UGP framework; it is presented here as a self-contained contribution. Readers interested in a candidate structural theory of the lepton masses within which Koide emerges are referred to [5]; readers interested only in the closed form and its cyclotomic identification can take the present paper as self-contained.

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Acknowledgments

AI assistants were used for adversarial review, coding assistance, and debugging during manuscript preparation and code development. All scientific claims, derivations, and numerical results were independently verified by deterministic scripts and Lean proofs; no AI-generated content appears as unverified scientific output.